

Informational Power and Institutional Design: When a Hegemon May Choose Consensus

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Abstract

When and why would a hegemon prefer consensus to majority rule in an international organization? I argue that consensus can be a technology of hegemonic power rather than a constraint upon it. In a formal model, a hegemon privately observes the value of cooperation and uses Bayesian persuasion to influence weaker states' entry decisions. Under majority rule, weak states can bypass the hegemon, eliminating any screening problem. Under unanimity, weak states must include the hegemon in every bargaining coalition without knowing its type, creating a screening cutoff at which their behavior changes discretely. This generates an upward jump in the hegemon's value function—a non-convexity that Bayesian persuasion exploits. The institutional comparison has a single-crossing property: there is a unique prior threshold, in closed form, above which the hegemon prefers unanimity and below which majority can dominate by making entry easier. The mechanism reframes institutional design as a trade-off between informational leverage under unanimity and easier participation under majority, providing a distributive explanation for consensus rules in organizations like the GATT/WTO.

1 Introduction

Most international organizations operate by consensus, even when powerful states drive their creation (Gould 2022). Why would a powerful state ever prefer consensus over majority rule, where decisions can pass with only a simple majority? This is a puzzle, as the international arena is where

material asymmetries—that is, differences in resources and influence among states—should matter most. If institutional rules reflect power, powerful states might be expected to favor arrangements that let them turn that power into a distributive advantage, meaning an ability to secure a greater share of benefits.

In bargaining, majority rule allows a powerful actor to form coalitions, exclude rivals, and keep the surplus (Baron and Ferejohn 1989; Kalandrakis 2006; Eraslan and Evdokimov 2019). Consensus does the opposite. By giving every participant a veto, it forces powerful states to surrender the very tools they might value most (Keohane 1984). Yet, some major international organizations, such as the GATT/WTO, are built around consensus (Steinberg 2002; Koremenos, Lipson, and Snidal 2001).

Existing answers do not fully resolve the puzzle. A common response claims that powerful states still influence outcomes through informal agenda power, even under consensus. Yet this answer is conceptually unsatisfying: under unanimity, all participants are veto players, so informal agenda influence cannot enable exclusion and surplus capture as majority rule does. More broadly, existing accounts either see consensus as constraining powerful states or as a concealment for conventional power, but they do not explain when and why a hegemon would prefer unanimity to a rule that allows exclusion.

This paper theorizes when and why a hegemon may prefer unanimity to majority rule in international organizations. The central claim is that consensus is not merely a constraint or procedural facade; rather, it is an institutional technology of power. Using Bayesian persuasion (Kamenica 2019), I show that consensus can make informational power more politically productive than the coalition arithmetic of majority rule. While the multi-receiver persuasion literature has examined information design under unanimity (Bardhi and Guo 2018), and Kim, Kim, and Van Weelden (2025) analyzes Bayesian persuasion in bilateral veto bargaining, neither offers the institutional comparison between voting rules that is central here. Thus, the key institutional choice is not between power and constraint, but between two distinct modes of exercising power.

Two forms of power are central to the analysis. *Agenda power* is the ability to control proposals, shape coalitions, and determine which alternatives reach the floor—the familiar Baron-Ferejohn logic in which the capacity to exclude is the source of surplus capture. *Informational power* is

the ability to shape beliefs about the value of cooperation; in international politics, large powers typically possess deeper technical expertise and greater ability to generate authoritative interpretations of proposed rules, whereas weaker states face limited analytical capacity. This paper's focus on information *design* under different voting rules complements the literature on information *aggregation*, where Feddersen and Pesendorfer (1998) show that unanimity is the worst rule for aggregating dispersed private information. The present model reverses the direction: information is concentrated in a single actor (the hegemon), and unanimity amplifies rather than impedes its strategic value (on veto structures and bargaining power more generally, see Diermeier and Myerson 1999). Under majority rule, agenda power dominates: weak states can exclude the hegemon, so beliefs affect expected values but create no strategic discontinuity. Under unanimity, informational power dominates: every actor becomes pivotal, and the hegemon's private information creates a screening problem that generates a non-convexity in its value function.

To formalize the argument, I develop a model in which a hegemon chooses between two institutional packages that differ only in the voting rule: majority and unanimity. Proposal rights are symmetric under both rules, isolating the effect of the voting rule itself. Before weaker states decide whether to enter, the hegemon privately observes the value of cooperation and can shape their beliefs about it. The central difference is that the two rules reward this informational advantage differently. Under majority rule, weak states can form coalitions without making the hegemon pivotal, eliminating the screening problem. Under unanimity, weak states must include the hegemon in every proposal and decide how much to offer without knowing the probability of acceptance. This screening generates a cutoff at which weak proposers switch from aggressive to conservative treatment of the hegemon, creating a discontinuity in the hegemon's value function that Bayesian persuasion exploits. This creates a persuasion opportunity absent under majority, although majority may still be preferred when it facilitates participation under less optimistic priors.

The analysis yields three main results. First, I characterize the screening cutoff under unanimity and show it has no analog under majority rule, where the hegemon's value function is linear in beliefs (Propositions 1–4). Second, I show that this screening creates an additional non-convexity in the hegemon's value function, providing a persuasion opportunity absent under majority (Proposition 5). Third, I establish that the institutional comparison has a single-crossing property: there is at most

one prior at which the ranking between unanimity and majority changes, and I provide a closed-form expression for the threshold (Lemma 1 and Theorem 1).

The substantive implication is that consensus is most valuable to a hegemon when the prospects of multilateral cooperation are promising enough that weaker states are willing to come to the table. Under those conditions, unanimity amplifies the hegemon's informational advantage: weaker states must include it in every deal without knowing what cooperation is truly worth, and the hegemon can shape their beliefs to exploit that uncertainty. When cooperation looks too uncertain for weaker states to participate at all, majority rule can be preferred—not because it gives the hegemon better bargaining terms, but because it lowers the bar for participation. These two forces produce a clean trade-off with a unique tipping point: above it, the hegemon prefers consensus; below it, it may prefer majority. The ranking never oscillates—the institutional comparison is not incomplete but conditional in a sharp and tractable way. Under identifiable informational conditions, consensus becomes an institutional form through which hegemonic power is exercised, not a constraint upon it.

In sum, this paper contributes to the study of international organizations in three ways. First, it offers a new mechanism explaining when and why a powerful state may rationally choose unanimity rather than just accept it. Second, it reframes institutional design as a trade-off between informational leverage under unanimity and easier participation under majority, rather than solely between power and constraint. Third, it provides a formal link between voting rules and the strategic value of private information.

The rest of the paper proceeds as follows. Section 2 presents a motivating example that previews the mechanism in a simplified three-player environment. Section 3 presents the general model. Sections 4 and 5 derive the bargaining logic under majority and unanimity, showing that unanimity creates a screening cutoff and a discrete jump in the hegemon's payoff that majority eliminates. Section 6 connects bargaining to entry and persuasion. Section 7 establishes the main result. Section 8 discusses extensions, the GATT/WTO application, and scope conditions. Section 9 concludes.

2 A Motivating Example

Before turning to the general model, consider a stripped-down environment with three players to build intuition for the mechanism. There is one hegemon H and two weak states W_1, W_2 . The value of cooperation is either $V(0) = 1$ (low) or $V(1) = 2$ (high); H observes the state, the weak states do not. Each weak state's outside option is 0; H 's outside option is $\alpha V(\theta)$ with $\alpha = 0.2$. A weak state is selected to make a take-it-or-leave-it offer specifying H 's share y_H ; if H rejects, each player receives their disagreement payoff.

Majority: exclusion eliminates screening. Under majority rule, W_1 and W_2 can pass a proposal without H 's vote. The proposing weak state excludes H from the coalition; H captures $0.2V(\theta)$ from its bilateral alternative regardless of what W believes. The hegemon's expected payoff is $0.2 + 0.2\mu$, where $\mu = \Pr(\theta = 1)$ —linear in beliefs, with no strategic discontinuity.

Unanimity: inclusion creates screening. Under unanimity, H 's vote is required. The proposing weak state must decide how much to offer H without knowing θ :

- *Aggressive offer* ($y_H = 0.2$): type $\theta = 0$ is indifferent and accepts; type $\theta = 1$ rejects (wants 0.4). W gets 0.8 if $\theta = 0$, gets 0 if $\theta = 1$.
- *Conservative offer* ($y_H = 0.4$): both types accept. W gets 0.6 if $\theta = 0$, gets 1.6 if $\theta = 1$.

The weak state prefers the aggressive offer when $0.8(1 - \mu) > 0.6(1 - \mu) + 1.6\mu$, which simplifies to $\mu < 1/9$. The screening cutoff $\mu^* = 1/9$ is the belief at which W switches from aggressive to conservative treatment of H .

The jump and overpayment. Below the cutoff, H 's expected payoff is the same as under majority: $0.2 + 0.2\mu$. Above the cutoff, H 's expected payoff jumps to 0.4—constant, independent of μ . The source is overpayment: on the conservative branch, type $\theta = 0$ receives 0.4 but would accept 0.2. Weak states cannot avoid this cost because they must buy H 's vote without knowing the type. The jump at $\mu^* = 1/9$ is 0.18, or about 16% of expected surplus at the cutoff.

Bayesian persuasion exploits the jump. The jump creates a non-convexity in H 's value function that Bayesian persuasion can exploit. Suppose the prior is $p < 1/9$. Without persuasion, H 's payoff under both rules is $0.2 + 0.2p$. But under unanimity, H can design a public signal that pushes W 's posterior to $1/9$ with probability $9p$ (inducing conservative behavior and payoff 0.4) and to 0 with

probability $1 - 9p$ (payoff 0.2). The resulting expected payoff is $0.2 + 1.8p$, strictly exceeding the majority payoff of $0.2 + 0.2p$ for every $p > 0$. Under majority, the payoff is already linear, so concavification adds nothing. This is the central asymmetry: unanimity creates a non-convexity that makes information strategically productive; majority does not.

Preview. The general model (Section 3 onwards) enriches this logic with N players, random proposer, two-round Baron-Ferejohn bargaining with discounting ($\beta < 1$), and an explicit entry stage with cost c . The richer environment produces two screening cutoffs (one per round), a richer non-convexity structure, and an entry threshold that creates a second channel through which the voting rule matters. Theorem 1 shows that the pattern here—unanimity dominates at sufficiently optimistic priors, majority can dominate only through easier entry at pessimistic priors—generalizes with a single-crossing property.

3 The Model

3.1 Environment

Definition 1 (Institutional Design Game). *The game $\Gamma(N, p, r, \alpha, \beta, c)$ has the following primitives.*

- *There are $N \geq 3$ players: one hegemon H and $N - 1$ weak states $W_1, \dots, W_{N-1}$.*
- *Nature draws $\theta \in \{0, 1\}$ with prior $\Pr(\theta = 1) = p \in (0, 1)$.*
- *The state determines the value of multilateral cooperation: $V(0) = 1$ and $V(1) = r > 1$.*
- *The hegemon observes θ ; the weak states do not.*
- *Disagreement payoffs: each W receives 0;¹ H receives $\alpha V(\theta)$, with $\alpha \in (0, 1/r)$.²*
- *The common discount factor is $\beta \in (0, 1)$.*

¹The normalization $d_W = 0$ is without loss of generality: it requires only that weak states' outside options are symmetric and strictly lower than the hegemon's. The pie $V(\theta)$ is then interpreted as the surplus above weak states' common outside option.

²The proportional form $d_H = \alpha V(\theta)$ captures bilateral alternatives that scale with the value of multilateral cooperation—the same factors that make multilateral cooperation valuable also improve the hegemon's outside option. The restriction $\alpha < 1/r$ ensures bilateral alternatives are strictly dominated by multilateral cooperation for all states of the world.

- *Each weak state that joins the institution pays an entry cost $c > 0$.*
- *The voting rule is $R \in \{M, U\}$: under majority (M), decisions pass with $q = \lfloor N/2 \rfloor + 1$ votes; under unanimity (U), decisions require all N votes.³ Both rules have symmetric proposal rights (each player recognized with probability $1/N$).*

Let $V_e(\mu) \equiv 1 + \mu(r - 1)$ denote the expected value of cooperation at posterior belief μ , and define $x \equiv (N - 1)\alpha r$ as a notational shorthand.

3.2 Timing

The game has three stages.

1. **Stage 0 (Institutional choice).** The hegemon chooses the voting rule $R \in \{M, U\}$.
2. **Stage 1 (Information and entry).** The hegemon commits to a public signal structure $\pi : \{0, 1\} \rightarrow \Delta(S)$. Nature draws θ ; the hegemon observes θ and the signal realization $s \in S$ is generated according to π . Weak states observe s , update to posterior $\mu = \Pr(\theta = 1 \mid s)$, and simultaneously decide whether to enter (paying cost c).
3. **Stage 2 (Bargaining).** If the institution forms, members play a two-round Baron-Ferejohn bargaining game over a pie of size $V(\theta)$:
 - **Round 1:** a proposer is selected uniformly at random (probability $1/N$). The proposer offers a division. All members vote according to rule R . If accepted, payoffs are realized. If rejected, the game proceeds to Round 2 with payoffs discounted by β .
 - **Round 2 (terminal):** a new proposer is selected uniformly at random. The proposer offers a division. All members vote. If accepted, payoffs are realized. If rejected, each player receives their disagreement payoff.

3.3 Solution concept

The solution concept is Perfect Bayesian Equilibrium (PBE). Strategies are sequentially rational given beliefs, and beliefs are updated by Bayes' rule on the equilibrium path. Off-path beliefs

³The model analyzes formal unanimity (every member has a veto). In practice, "consensus" at the WTO refers to the norm of not calling a formal vote, which is distinct from a formal unanimity requirement. The results apply most directly to settings where each member has an effective veto, whether de jure or de facto.

satisfy sequential rationality.⁴

3.4 Information structure

Throughout the game, the hegemon's information sets are singletons: H knows θ at every decision node. Each weak state's information sets pool the two branches of θ : at any decision node, W_i knows the public signal realization and the history of proposals and responses, but not θ . After the public signal, weak states hold posterior μ over θ . This asymmetry is what generates the screening problem when a weak state proposes under unanimity.

Figures 1 and 2 present the extensive form of the game: Figure~1 covers institutional choice, Bayesian persuasion, and entry (Stages 0–1); Figure~2 details the bargaining subgame under unanimity (Stage 2).

⁴Tie-breaking convention: when H is indifferent between accepting and rejecting an offer (offer equals reservation value), H accepts.

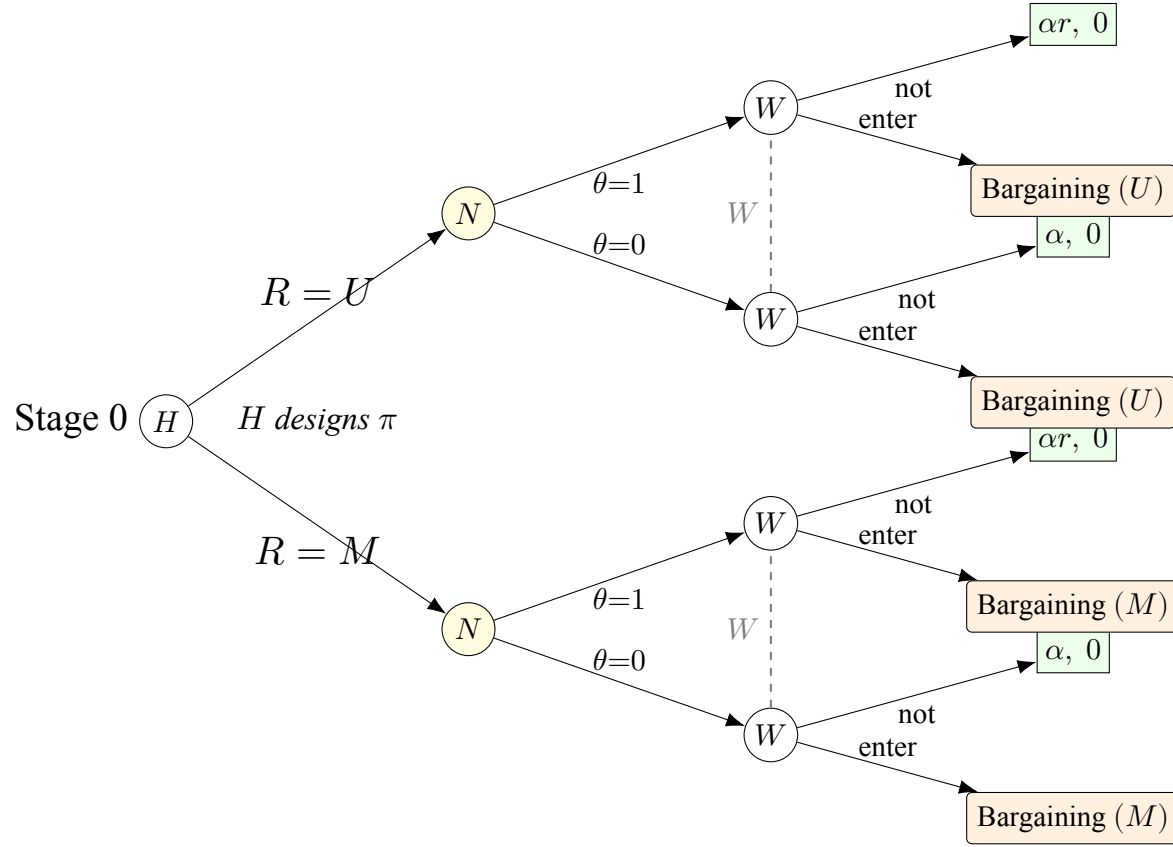


Figure 1: Extensive form: Stages 0–1 (institutional choice, information design, and entry). At Stage 0, H selects voting rule $R \in \{M, U\}$ and commits to a Bayesian persuasion signal π . Nature draws $\theta \in \{0, 1\}$; weak states observe signal realization s , form posterior $\mu = \Pr(\theta=1 \mid s)$, and decide entry (cost c). Dashed lines: W 's information sets (W does not observe θ). Non-entry payoffs: $(u_H, u_W) = (\alpha V(\theta), 0)$.

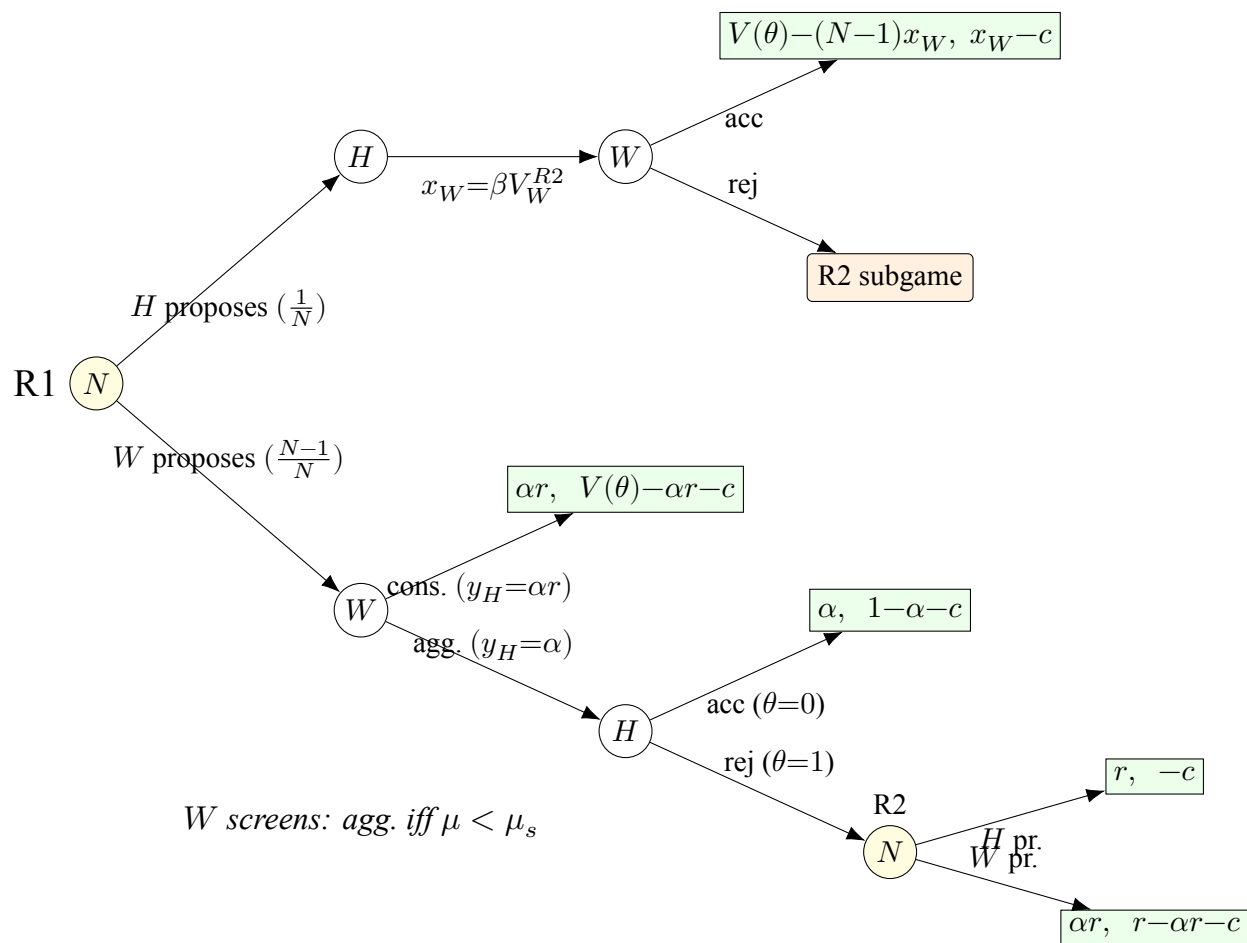


Figure 2: Extensive form: Stage 2 (bargaining under unanimity, Rounds 1–2). Payoffs are (u_H, u_W) . A proposer is selected uniformly ($1/N$). When W proposes, W chooses between aggressive ($y_H = \alpha$) and conservative ($y_H = \alpha r$). Under the aggressive offer, type $\theta=0$ accepts (tie-breaking) and $\theta=1$ rejects, revealing the state and leading to R2 (payoffs discounted by β). In R2, W knows $\theta=1$ (complete information); W offers αr , H offers 0; disagreement yields $(\alpha r, -c)$. When H proposes in R1, H offers βV_W^{R2} to each W ; W accepts in equilibrium. Under majority rule, H 's vote is not pivotal, eliminating the screening mechanism (Proposition 1).

3.5 Preview of the main result

The paper’s central result (Theorem 1) establishes that the institutional comparison has the *single-crossing property*: there exists at most one prior p^* such that the hegemon prefers unanimity for $p > p^*$ and majority for $p < p^*$. The mechanism rests on three building blocks developed in Sections 4–@ref(entry_bp): (1) majority produces no screening, so the hegemon’s payoff is linear in beliefs (Section 4); (2) unanimity forces weak states to screen the hegemon’s type, creating a discrete upward jump in the hegemon’s expected payoff (Section 5); and (3) this jump generates non-convexity that Bayesian persuasion exploits, more than compensating for the harder entry conditions under unanimity (Section @ref(entry_bp)). The following sections develop each building block before assembling them into the institutional comparison in Section 7.

4 Majority Rule: Linear Bargaining Without Screening

Under majority rule, a proposal passes with $q = \lfloor N/2 \rfloor + 1$ votes. The key structural feature is that weak states can form a winning coalition without including the hegemon. When a weak state proposes, it assembles a majority from the other weak states alone; the hegemon, excluded from the coalition, captures its bilateral alternative $\alpha V(\theta)$ regardless of what weak states believe about the state of the world.

Proposition 1 (Majority rule produces no screening). *Under majority rule with symmetric proposal rights, the hegemon’s expected continuation value $E_\theta[V_H^{R1}(\theta, \mu, M)]$ is affine in posterior beliefs μ . There is no screening cutoff.*

Proof. See Appendix B.1. □

Because the hegemon’s vote is never needed, weak states never face a choice between offers that depend on inferring the hegemon’s type. The hegemon’s private information affects the value of the agreement but creates no strategic discontinuity. Majority transforms bargaining into coalition arithmetic: the proposer buys the cheapest votes, and the hegemon is simply paid its outside option. Bayesian persuasion under majority therefore operates only through the entry channel—it can induce weak states to participate, but it cannot extract screening rents from the bargaining process

itself.

5 Consensus: Screening Through Pivotal Inclusion

Under unanimity, every proposal requires the approval of all members, including the hegemon. When a weak state proposes, it must secure H 's vote without knowing whether the pie is high or low. As in the motivating example (Section 2), this creates a screening problem: the proposer must choose between an aggressive offer that only the low type accepts and a conservative offer that both types accept. The conservative offer overpays the low type—the price of uncertainty.

This screening logic operates in both bargaining rounds. In each round, the weak proposer's indifference between aggressive and conservative offers defines a screening cutoff: a belief threshold above which the proposer switches from aggressive to conservative treatment. The cutoffs have closed-form expressions (Appendix A).

Proposition 2 (R1 screening cutoff under consensus). *Under unanimity, there exists a unique screening cutoff $\mu_s^{R1} \in (0, 1)$ such that the weak proposer prefers the aggressive offer for $\mu < \mu_s^{R1}$ and the conservative offer for $\mu > \mu_s^{R1}$. In the principal regime, the cutoff is:*

$$\mu_s^{R1} = \frac{\phi - \sqrt{\phi^2 - 4(N-2)}}{2(N-2)}, \quad \phi \equiv \frac{rN - \beta(N-1+r)}{\beta(r-1)}. \quad (1)$$

In this regime, the cutoff does not depend on α . For $\beta = 1$, it simplifies to $\mu_s^{R1} = 1/(N-2)$.

Proof. See Appendix B.2. □

The independence of the screening cutoff from the hegemon's outside option strength is substantively important. What determines how aggressively weak states treat the hegemon is not how valuable bilateral alternatives are, but how consequential the hegemon's private information is for the value of cooperation, how patient the bargainers are, and how many states must agree.

Proposition 3 (Screening creates an informational rent). *Under unanimity, the hegemon's expected*

continuation payoff in R1 has an upward jump at μ_s^{R1} :

$$Jump = (1 - \mu_s^{R1}) \cdot \frac{(N - 1)\beta(r - 1)}{N^2} > 0. \quad (2)$$

Proof. See Appendix B.3. □

At the screening cutoff, weak proposers switch from treating the hegemon as the low type to hedging against the possibility of the high type. The conservative offer pays the low type as if it could be the high type, creating a discrete upward shift in the hegemon's expected payoff. The screening rent is largest when the hegemon's private information is most consequential for the value of cooperation (large r), when bargaining is patient enough to make the continuation threat credible (large β), and in moderately-sized organizations—large enough that the hegemon's vote matters, but not so large that the cost of unanimous agreement dilutes the per-member advantage ($(N - 1)/N^2$ peaks at intermediate N).

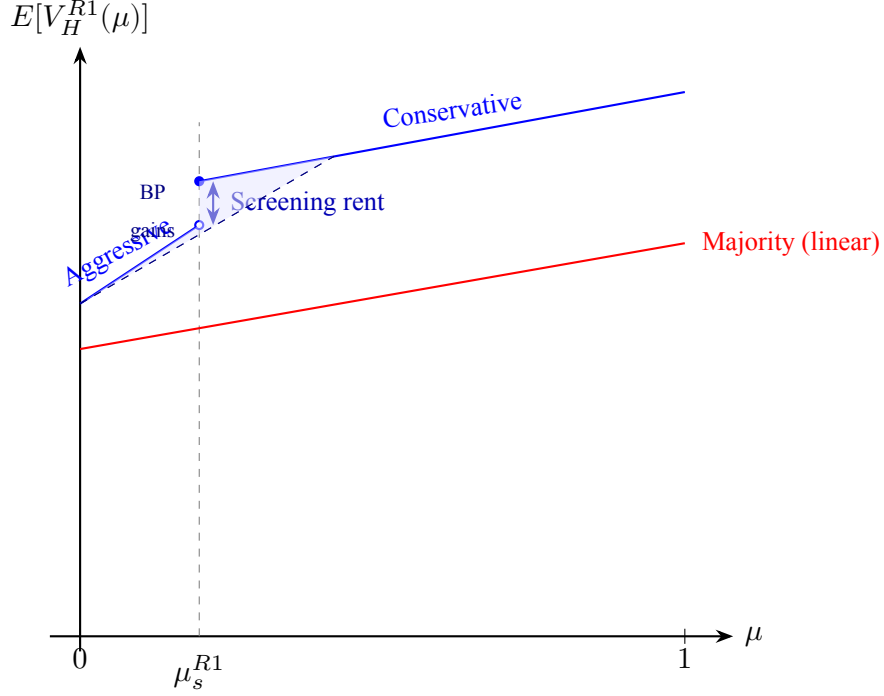


Figure 3: Schematic of the hegemon's expected R1 payoff as a function of posterior belief μ , based on Example 1 parameters ($N = 5$, $r = 1.5$, $\alpha = 0.3$, $\beta = 0.9$). Under majority (red), the payoff is linear in μ with no discontinuity. Under unanimity (blue), the payoff has two branches separated by an upward jump at the screening cutoff $\mu_s^{R1} \approx 0.197$: to the left, weak proposers play aggressively; to the right, they switch to conservative offers that overpay the low type. The shaded region shows where the concave envelope (dashed) lies above the value function, creating gains from Bayesian persuasion.

Example 1 (Screening jump: concrete magnitudes). Let $N = 5$, $r = 1.5$, $\alpha = 0.3$, $\beta = 0.9$. The R1 screening cutoff is $\mu_s^{R1} \approx 0.197$. At beliefs just below the cutoff, weak proposers play aggressively and the hegemon's expected payoff is $E[V_H^{R1}] \approx 0.544$. Just above the cutoff, the payoff jumps to ≈ 0.602 —an increase of about 5.3% of expected surplus. Under majority at the same belief, $E[V_H^{R1}] \approx 0.428$, with no jump anywhere. The screening mechanism gives the hegemon 27% more than majority on the aggressive branch and 41% more on the conservative branch. The discrete gap between branches is the non-convexity that Bayesian persuasion exploits.

6 Entry and Bayesian Persuasion

A weak state enters the institution if its expected bargaining payoff exceeds the entry cost $c > 0$.⁵ Because unanimity gives the hegemon a higher payoff conditional on participation (as established in Section 7), it leaves less surplus for weak states, making entry harder: $\tau(U) \geq \tau(M)$.

The hegemon's reduced-form value function is $v(\mu, R) = E_\theta[V_H^{R1}(\theta, \mu, R)]$ when weak states enter and zero otherwise. Under majority, this function is affine above the entry threshold—a single non-convexity. Under unanimity, the screening jump at μ_s^{R1} creates an additional non-convexity.

Proposition 4 (Unanimity creates an additional persuasion opportunity). *The value function $v(\mu, U)$ has a screening non-convexity at μ_s^{R1} that is absent in $v(\mu, M)$.*

Proof. See Appendix B.4. □

By the standard Bayesian persuasion result (Kamenica and Gentzkow 2011), the hegemon's optimal payoff under rule R is $\Pi_H^*(R) = \text{cav } v(p, R)$ —the concavification of the value function evaluated at the prior. Under majority, concavification exploits only the entry threshold: the hegemon can induce participation but cannot extract screening rents. Under unanimity, concavification can exploit both entry and screening, inducing posteriors that straddle μ_s^{R1} and place mass on both sides of the jump. This dual exploitation is the paper's distinctive contribution. Figure ?? illustrates both value functions and their concavifications.

7 Institutional Choice

This section establishes the paper's main result: conditions under which the hegemon prefers unanimity.

⁵The entry cost c may represent fixed costs of negotiating accession, per-round costs of preparing delegations, or political costs of committing to multilateral disciplines. In established organizations, c is best interpreted as the cost of *substantive engagement*: allocating scarce diplomatic and analytical resources to a particular negotiating round.

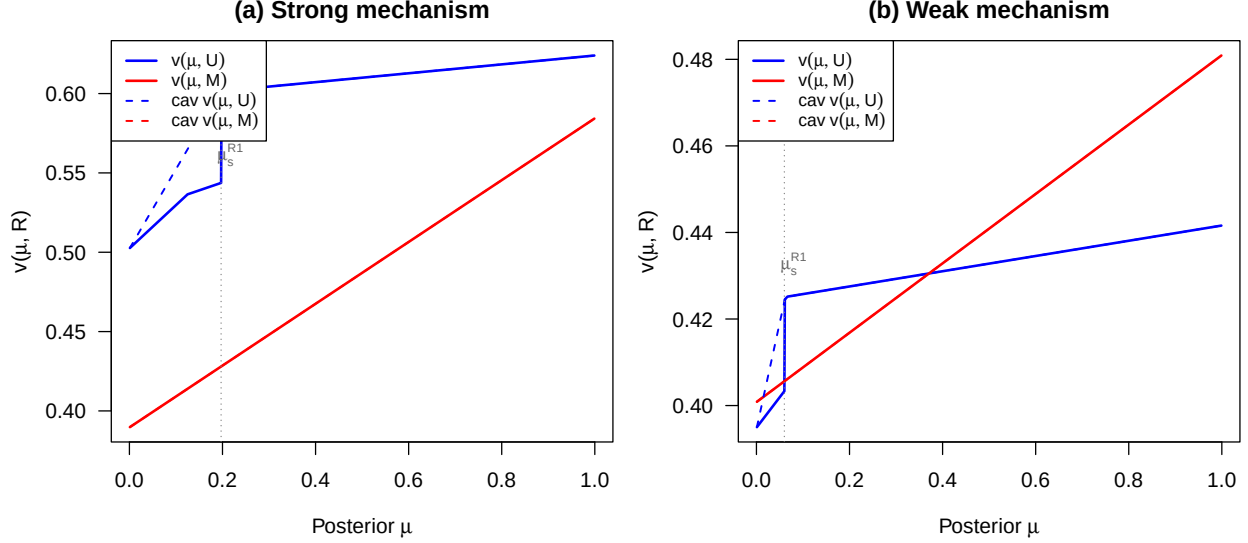


Figure 4: (#fig:bp_illustration) Value functions and concavification under majority (red) and unanimity (blue). Dashed lines show concave envelopes. Left panel (a): baseline parameters ($N = 5$, $r = 1.5$, $\alpha = 0.3$, $\beta = 0.9$, $c = 0.1$) where the screening jump under unanimity creates a large concavification gain. Right panel (b): parameters near the boundary of the unanimity-preferred region ($N = 5$, $r = 1.2$, $\alpha = 0.3$, $\beta = 0.7$, $c = 0.1$) where the mechanism is weaker and the concavification gains are smaller.

7.1 Conditional payoff comparison

Before turning to the full institutional comparison with persuasion, I establish a result that is central to the analysis: conditional on the institution forming, unanimity gives the hegemon a strictly higher expected continuation payoff than majority for all interior beliefs.

Lemma 1 (Conditional payoff dominance of unanimity). *Define the threshold*

$$\alpha^*(N, \beta) \equiv \frac{\beta(q-1)}{N(N-1) - \beta(N^2 - N - q + 1)}, \quad q = \lfloor N/2 \rfloor + 1.$$

For $\alpha < \alpha^*(N, \beta)$ and every $\mu \in (0, 1]$,

$$E_\theta[V_H^{R1}(\theta, \mu, U)] > E_\theta[V_H^{R1}(\theta, \mu, M)].$$

Proof. See Appendix B.5. □

The lemma says that, conditional on the institution forming, the hegemon always does better un-

der unanimity—provided its bilateral alternatives are not too strong. The intuition has two parts. When the hegemon proposes, unanimity depresses weak states' continuation values (because rejection leads to a game where the hegemon is pivotal), allowing the hegemon to buy agreement more cheaply. When a weak state proposes, unanimity forces the proposer to secure the hegemon's vote, generating the screening rent identified in Proposition 3. Both forces favor the hegemon. The condition $\alpha < \alpha^*$ ensures that the cost of buying unanimous agreement does not offset these advantages. As the organization grows, the threshold tightens: in larger organizations, unanimity is advantageous only when the hegemon's bilateral alternatives are modest relative to multilateral cooperation.

An immediate corollary is that entry is always harder under unanimity ($\tau(U) \geq \tau(M)$): because the hegemon captures more surplus, less is available for weak states. Without persuasion, the institutional comparison is therefore ambiguous: unanimity dominates when the institution forms under both rules, but majority dominates when only it can induce entry.

7.2 The main result

With Bayesian persuasion, the hegemon's payoff under each rule is $\Pi_H^*(R) = \text{cav } v(p, R)$. The institutional choice reduces to comparing these concavified values.

Theorem 1 (Threshold prior for institutional choice). *Suppose $\alpha < \alpha^*(N, \beta)$. The institutional comparison has the single-crossing property: there exists at most one prior p^* at which the ranking between unanimity and majority changes. Specifically:*

- (a) *When entry costs are low enough that the institution forms under unanimity at all priors, unanimity strictly dominates for every $p \in (0, 1)$.*
- (b) *When entry occurs under majority but not unanimity at low priors, there exists a unique threshold p^* such that majority dominates for $p < p^*$ and unanimity dominates for $p > p^*$.*

The threshold p^ has a closed-form expression (Appendix B.6). In all cases, unanimity dominates at sufficiently optimistic priors, and any advantage of majority arises exclusively through the entry margin.*

Proof. See Appendix B.6. □

The theorem resolves the institutional choice puzzle. Majority's only advantage is that it makes participation cheaper for weak states, because the hegemon captures less surplus. But this advantage is relevant only when the prior is pessimistic enough that entry is the binding constraint. At optimistic priors—where the value of cooperation is likely high and the institution would form under either rule—unanimity strictly dominates because the screening mechanism gives the hegemon an informational rent that majority cannot replicate.

The single-crossing property means there is no oscillation: the ranking switches at most once. Consensus is not sometimes better and sometimes worse depending on delicate parameter configurations; it is better whenever the institution is viable.

Consensus is most valuable when the hegemon's private information is most consequential for the value of cooperation, when bargaining is patient, and in moderately-sized organizations. The advantage of unanimity increases when the screening rent is large relative to the linear payoff under majority—that is, when informational power is a potent substitute for agenda power.

7.3 Numerical characterization

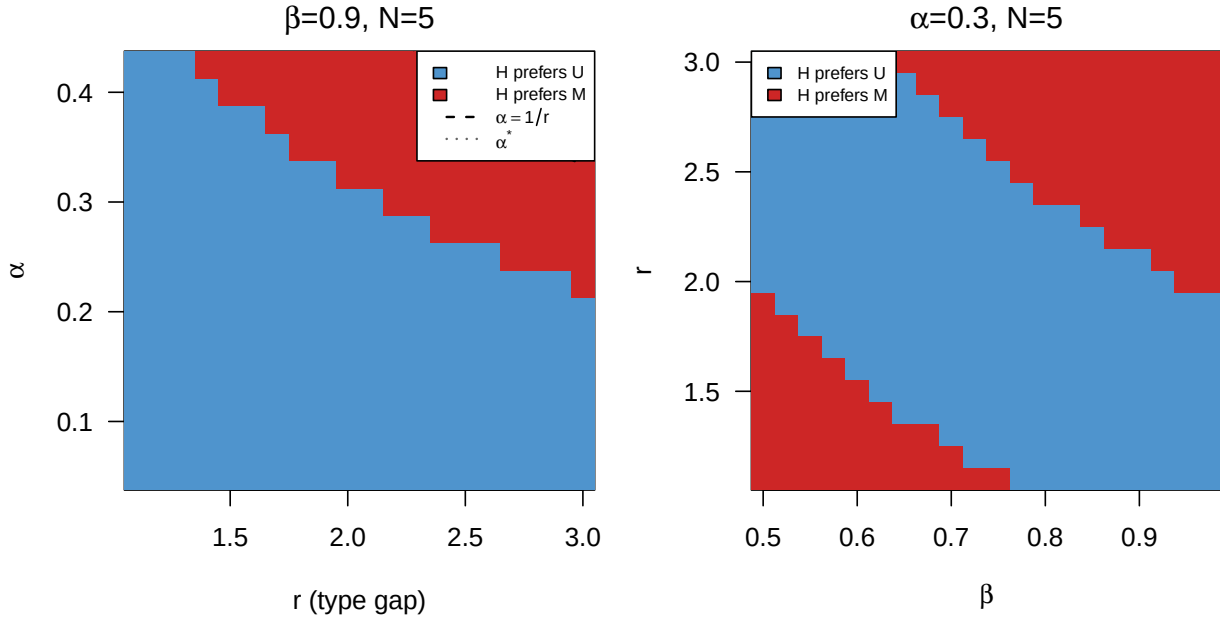


Figure 5: (#fig:parameter_regions)Parameter regions where the hegemon prefers unanimity (blue) vs majority (red) at prior $p = 0.05$, restricted to $\alpha < \alpha^*(N, \beta)$ (the domain where Lemma 1 applies). Left: r vs α plane with $\beta = 0.9$, $N = 5$, $c = 0.1$. Right: β vs r plane with $\alpha = 0.3$, $N = 5$, $c = 0.1$.

Figure ?? shows that the hegemon prefers unanimity across a broad range of parameters when the prior is low ($p = 0.05$). In the blue regions, unanimity dominates globally—case (a) of Theorem 1. The unanimity-dominance region is larger when the hegemon’s private information is more consequential (higher r) and when bargaining is more patient (higher β). In the red regions, case (b) applies: the threshold p^* lies above 0.05, so majority dominates at this pessimistic prior, but unanimity would dominate at higher priors.

8 Discussion

8.1 Partial Agenda Influence and Informal Power

The baseline model uses symmetric proposal rights ($\rho_H = 1/N$) to isolate the effect of the voting rule. A natural objection is that powerful states exercise informal agenda influence—through control over drafting, chairing of meetings, or greater diplomatic salience—so the symmetric baseline is unrealistic. Appendix C extends the model to allow H to be recognized with probability $\rho_H > 1/N$, while each W is recognized with probability $\rho_W = (1 - \rho_H)/(N - 1)$.

The extension reveals that consensus and informal agenda influence are not substitutes—they interact non-monotonically. Under majority, the value function remains linear in beliefs regardless of ρ_H : no screening, no jump. Under unanimity, the screening cutoff μ_s^{R2} is unchanged, but the jump magnitude becomes $(1 - \mu_s^{R2})(1 - \rho_H)\alpha(r - 1)$, decreasing in ρ_H . In R1, three effects operate simultaneously: a frequency effect (W proposes less often, reducing screening opportunities), a gap effect (H ’s higher continuation value widens the spread between aggressive and conservative offers), and a cutoff effect (the screening threshold rises). The net R1 jump scales with $\rho_H(1 - \rho_H)$ holding the cutoff fixed, producing a hump-shaped relationship.

Proposition 5 (Non-monotonic effect of agenda influence). *Under unanimity, the R1 screening jump as a function of recognition probability ρ_H is:*

$$J(\rho_H) = (1 - \mu_s^{R1}(\rho_H))(1 - \rho_H)\rho_H\beta(r - 1).$$

Holding μ_s^{R1} fixed, J is maximized at $\rho_H = 1/2$. In the full model, μ_s^{R1} increases with ρ_H .

shifting the peak below $1/2$. Symmetric proposal rights ($\rho_H = 1/N$) can therefore be understood as participation-preserving restraint rather than as absence of hegemonic influence.

8.2 Illustration: The GATT/WTO

The GATT/WTO illustrates the interaction between consensus and informal agenda influence. Since 1947, the multilateral trading system has operated predominantly by consensus, even though formal voting provisions exist (Steinberg 2002). The United States and other major trading powers were central to the system's creation and could plausibly have imposed majority rule.

The model's informational asymmetry assumption finds support in the trade context. Evaluating the effects of tariff schedules, anticipating distributive consequences, and assessing long-term implications of rule changes require sustained investment in analytical infrastructure. Major trading powers maintain large permanent delegations with deep sectoral expertise; many developing country delegations operate with far smaller teams, particularly during complex technical phases of negotiations. This capacity asymmetry means that weaker states often face genuine uncertainty about the value of proposed agreements.

In practice, major trading powers exercise informal agenda influence through the Green Room process, control over drafting, and technical expertise. The partial agenda influence extension provides the framework for interpreting this: moderate informal influence can amplify the screening channel—it increases the stakes when W does get to propose—while monopoly agenda control would eliminate participation, destroying the institution. The persistence of consensus is consistent with the coexistence of informal influence and formal equality.

The model's mechanism explains a pattern that existing accounts describe but do not fully explain: powerful states in the WTO appear to benefit from consensus despite lacking formal agenda control. The mechanism is not merely that informal practices survive consensus, but that consensus creates the strategic environment—pivotal inclusion under uncertainty—in which informational power becomes most valuable. The model does not claim that the GATT/WTO was designed for this reason, or that all aspects of WTO politics reduce to this mechanism. It shows that consensus and informational asymmetry interact in a way that can benefit powerful states, offering one explanation for why hegemonic preference for consensus is not paradoxical.

The informational mechanism generates a distinctive observable implication: the hegemon’s advantage from consensus should be largest in negotiations where informational asymmetry is substantial and where weak states lack independent analytical capacity. In the WTO context, this suggests that consensus should matter most in complex regulatory areas—services, intellectual property, investment—where evaluating proposals requires deep technical expertise, and least in areas where distributive consequences are transparent, such as linear tariff cuts. This prediction distinguishes the informational mechanism from legitimacy accounts (which predict consensus matters uniformly) and from self-enforcement accounts (which predict consensus matters most when enforcement is weak, regardless of information).⁶

The GATT-to-WTO transition. The model’s institutional comparison offers a lens for understanding the evolution of decision-making rules in the multilateral trading system. The GATT 1947 provided for formal majority voting—simple majority for joint action, two-thirds for amendments—and voting was used regularly in the organization’s early years. Beginning in the late 1950s, however, consensus gradually displaced voting as the operative decision-making norm; since 1959, virtually all GATT/WTO legislative decisions have been taken by consensus (Steinberg 2002, 344). The transition coincided with the *en masse* accession of developing countries, which fundamentally changed both the size and the informational composition of the membership.

The model suggests a structural interpretation. In the early GATT, membership was small and relatively homogeneous—predominantly industrialized economies with similar analytical capacities. In such an environment, informational asymmetry between the hegemon and other members was modest, and the primary challenge was inducing participation in a new and untested institution. These are precisely the conditions under which Theorem 1 predicts majority dominance: when forming the institution is the binding constraint (low prior, high effective entry cost), the easier participation conditions under majority outweigh the screening advantages of unanimity. As the membership expanded and diversified, two shifts occurred simultaneously: the organization became more viable

⁶The $K > 2$ extension in Appendix D provides a formal basis for this implication. As the number of relevant states increases—corresponding to negotiations that bundle multiple issue dimensions under a Single Undertaking (intellectual property, services, industrial goods, agriculture)—the parametric threshold α_3^* tightens: the hegemon needs a weaker outside option for unanimity to dominate. In the model, higher K dilutes informational power because the screening problem becomes more complex, with correction terms that can work against H (see Section D.5). This is consistent with the contrast between focused early GATT rounds (few issue dimensions, K effectively small, consensus effective) and the Doha Round (many dimensions bundled via Single Undertaking, K large, consensus stalled). The erosion is not merely about weaker states acquiring expertise—it is also about the dimensionality of the agenda itself.

(lowering the effective entry barrier), and the informational gap between the hegemon and newer members widened (increasing r). Both shifts move the institutional comparison toward unanimity. The consensus norm that emerged was not merely a concession to the principle of sovereign equality—it was a rational adaptation to an environment in which informational power had become the hegemon’s most valuable asset.

More broadly, the model predicts that the transition from majority to consensus should be most pronounced in *new regimes*—institutional settings where the benefits of cooperation are genuinely uncertain and where participants lack prior experience to assess the value of proposed agreements. When a new international organization operates within an already-established regime (with known templates, precedents, and track records), the effective uncertainty about the value of cooperation is low regardless of the formal novelty of the institution. It is the novelty of the *regime*, not the organization, that generates the high entry costs and informational asymmetries under which majority initially dominates. The early GATT exemplifies this: multilateral trade liberalization was an untested project with no historical precedent, and the informational demands of evaluating tariff schedules, anticipating distributive consequences, and assessing long-term welfare effects were substantial. These conditions made majority rule the natural starting point and consensus the natural destination.

8.3 Scope

Why symmetric proposal rights? Under either voting rule, if the hegemon were the exclusive proposer and weak states had no outside option from bargaining ($d_W = 0$), weak states’ expected bargaining payoff would be zero. No weak state would pay an entry cost $c > 0$, and the institution would never form. Monopoly proposal power is self-defeating when participation is voluntary. The relevant institutional choice is therefore between voting rules, holding proposal rights symmetric. Other forms of agenda influence—such as greater recognition probability—are discussed in Section 8.1.

Why not compare consensus with hegemonic agenda control? Because agenda control combined with any participation constraint eliminates the institution itself. The puzzle is not why a hegemon does not impose dictatorship, but why it would choose a rule that makes its own approval necessary

when majority would allow others to bypass it.

When consensus favors the hegemon. The mechanism requires that: (i) informational asymmetry is relevant (H knows θ , W does not); (ii) the hegemon has a valuable outside option ($\alpha > 0$); (iii) entry is costly and belief-dependent; (iv) the voting rule makes H pivotal.

The model assumes the hegemon can commit to an information structure before observing θ . This is a strong assumption that warrants discussion. Three considerations support its plausibility in the IO context. First, *reputation*: in repeated interactions, a hegemon that deviates from an announced disclosure policy suffers credibility costs in future negotiations, which can sustain commitment in equilibrium. Second, *institutional transparency rules*: organizations like the WTO have formal provisions for notifications, trade policy reviews, and dispute settlement that constrain and channel information disclosure, serving as partial commitment devices. Third, *upper bound interpretation*: the Bayesian persuasion payoff represents the maximum value of the hegemon's informational advantage. Even if full commitment is unavailable, the qualitative institutional comparison—unanimity creates screening opportunities that majority does not—may survive under weaker information structures, such as evidence games (Glazer and Rubinstein 2004) or verifiable disclosure (Milgrom 1981).

When majority dominates. By Lemma 1, for $\alpha < \alpha^*(N, \beta)$, majority can never dominate through a higher conditional-on-entry payoff. Any advantage must come through the entry margin: $\tau(U) > \tau(M)$. This is more likely when entry costs are high (creating a large gap where the institution forms only under majority), when α approaches α^* (weakening the conditional dominance), or when information is approximately public. For $\alpha \geq \alpha^*$, majority can also dominate through conditional bargaining payoffs near $\mu = 1$, because unanimity's cost of buying $N - 1$ votes outweighs the screening advantage.

Figure ?? plots $\alpha^*(N, \beta)$ as a function of organization size for three levels of patience. The threshold decreases with N because unanimity requires buying $N - 1$ votes, while majority requires only $q - 1$. For large organizations ($N > 50$), α^* becomes small: the conditional dominance result applies only when the hegemon's bilateral alternative is modest relative to multilateral cooperation. This does not eliminate the mechanism: even when Lemma 1 fails globally, the screening jump at μ_s^{R1} persists, and Bayesian persuasion can still exploit it at low priors. The condition restricts when

unanimity dominates *everywhere*, not when it dominates *somewhere*.

The decreasing relationship between α^* and N has a substantive implication. A hegemon whose bilateral alternatives are moderate relative to multilateral cooperation— α well below α^* —benefits from consensus. But as bilateral alternatives strengthen, the screening mechanism no longer compensates for the cost of buying unanimous agreement. The proliferation of preferential trade agreements can be understood, in the model's terms, as an increase in α : bilateral and regional deals strengthen the hegemon's outside option relative to multilateral cooperation (Bhagwati 2008). As α rises past α^* , the hegemon's institutional preference shifts away from consensus—not because multilateral cooperation is less valuable, but because the hegemon can capture a larger share of that value bilaterally. This is consistent with the observation that hegemonic support for consensus-based institutions tends to weaken precisely when bilateral alternatives become more attractive.

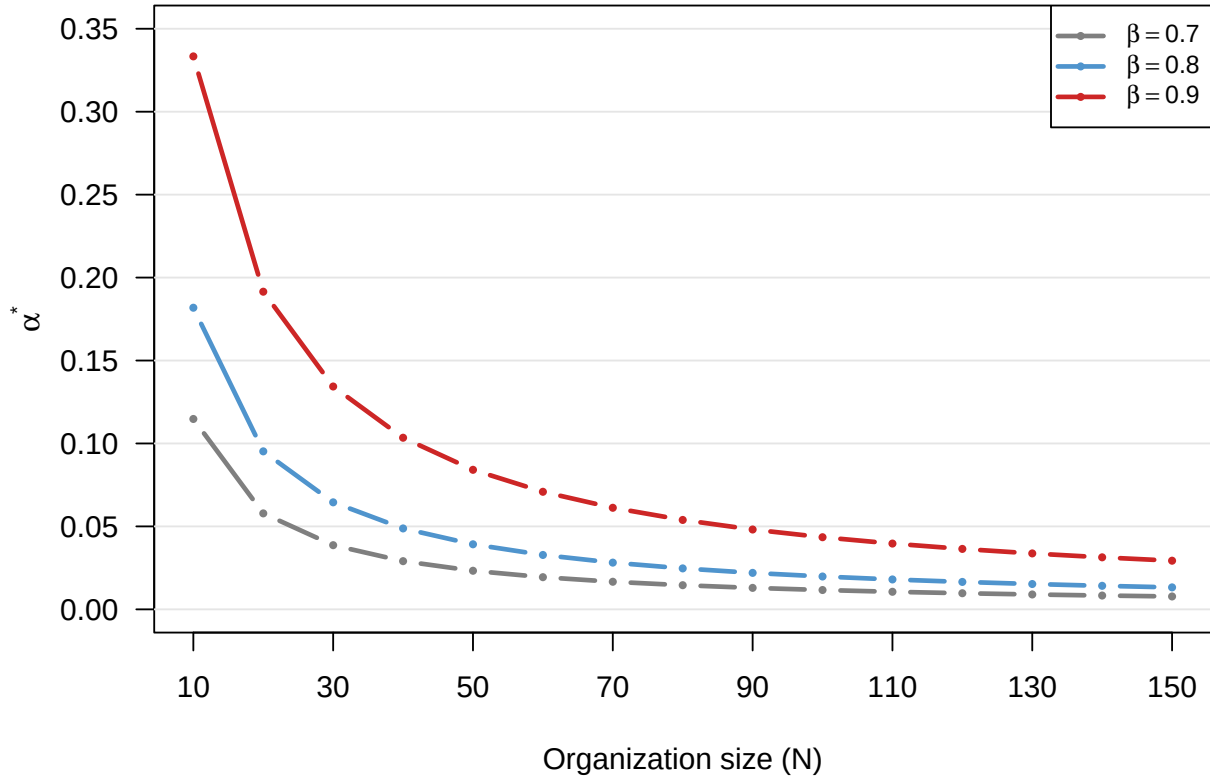


Figure 6: Threshold $\alpha^*(N, \beta)$ for conditional payoff dominance (Lemma 1) as a function of organization size N , for three discount factors. Below each curve, unanimity gives the hegemon a strictly higher continuation payoff than majority for all beliefs. The threshold decreases with N : in larger organizations, unanimity's vote-buying cost grows relative to the screening advantage, so the mechanism requires a weaker bilateral alternative.

For $\alpha \geq \alpha^*$, the conditional payoff ranking is ambiguous: unanimity can dominate at low μ (where the screening advantage is strong) while majority dominates near $\mu = 1$ (where the vote-buying cost is decisive). In this regime, the institutional comparison depends on the prior p and cannot be resolved by a single lemma. The paper's main results are stated for $\alpha < \alpha^*$, which ensures a clean separation between the conditional and entry channels.

Alternative explanations. The informational mechanism is complementary to, not a substitute for, other rationalist explanations of consensus. Legitimacy and compliance accounts emphasize that consent enhances durability of agreements, but address why agreements are honored rather than why a hegemon benefits from the bargaining structure. Self-enforcing agreement models (Maggi and Morelli 2006) show that unanimity can be optimal when enforcement is weak, but apply primarily to symmetric states and collective action problems rather than to asymmetric players with private information. Issue linkage accounts explain why broad agendas favor consensus but not why a hegemon with superior information would prefer unanimity in a single-issue environment. This paper isolates a distributive-informational mechanism within the rationalist framework common to these accounts, in the spirit of Fearon (1995).

Broader implications. Institutions do not merely reflect power; they select which form of power becomes politically productive. Majority rule rewards coalition arithmetic; unanimity rewards informational influence. This reframing implies a dynamic: as weaker states invest in independent analytical capacity, the informational asymmetry narrows, the returns to persuasion decline, and the hegemon's institutional incentive to prefer consensus should weaken. These dynamic implications are suggestive extrapolations from the static mechanism, not formal predictions of the model. A proper treatment would require a repeated game with endogenous information acquisition, which is beyond the scope of this paper. The value of consensus to a hegemon should covary with the distribution of technical expertise across the membership.

9 Conclusion

This paper has shown when and why consensus can be a rational choice for a hegemon. The key is to distinguish between two modes of exercising power under different voting rules. Majority rule with symmetric proposals lets weak states bypass the hegemon, eliminating the screening problem.

Unanimity forces weak states to include the hegemon under uncertainty, creating a screening cutoff that generates a non-convexity in the hegemon's value function. Bayesian persuasion exploits that non-convexity. The institutional comparison reduces to a single threshold: above it, unanimity dominates; below it, majority can dominate through easier entry. The ranking never oscillates.

The argument does not imply that all consensus institutions are designed for this reason, or that informal power disappears under consensus. It shows something more specific: under identifiable conditions on informational asymmetry and the prospects of cooperation, unanimity becomes the preferred institutional arrangement for a hegemon. Consensus may be less a concession by power than a way of exercising power through a different channel—the channel of information.

The model has a few limitations. The mechanism depends on a binary type space, which produces a discrete screening cutoff and an upward jump in the hegemon's value function. With a finite number of types $K > 2$, the screening problem would generate $K - 1$ cutoffs. I conjecture that each cutoff produces an upward jump in the hegemon's value function, so that the non-convexity structure becomes richer rather than weaker—but this remains to be verified for the specific payoff structure of this model. However, with a continuous type space, the weak proposer's offer schedule would generically be smooth, eliminating the discrete jumps that Bayesian persuasion exploits. That said, discrete types are substantively natural in this setting: states typically evaluate multilateral cooperation in terms of qualitatively distinct regimes (e.g., ambitious vs. modest trade rounds) rather than marginal variations in value. The mechanism is therefore robust to richer finite type spaces but may not survive the passage to the continuum. Whether informational power retains strategic value under continuous types—perhaps through different channels than the screening non-convexity identified here—is an open question.

Several extensions remain. First, if weaker states can invest in independent analytical capacity, informational power erodes endogenously over time. Second, heterogeneous outside options across weak states (middle powers) would enrich the screening dynamics. Third, repeated play within an established consensus institution could generate learning that narrows the informational gap. These directions point toward a richer theory of how informational power rises and declines within institutional frameworks.

Appendix A: Bargaining Derivations

Notation summary

Symbol	Meaning	Defined in
N	Number of players (1 hegemon + $N - 1$ weak states)	Def. 1
$\theta \in \{0, 1\}$	State of the world	Def. 1
p	Prior $\Pr(\theta = 1)$	Def. 1
μ	Posterior belief $\Pr(\theta = 1 \mid s)$	Sec. 4
$V(\theta)$	Value of cooperation: $V(0) = 1, V(1) = r$	Def. 1
$r > 1$	High-type value	Def. 1
$\alpha \in (0, 1/r)$	Outside-option share: $d_H = \alpha V(\theta)$	Def. 1
$\beta \in (0, 1)$	Common discount factor	Def. 1
$c > 0$	Entry cost per weak state	Def. 1
$V_e(\mu)$	Expected cooperation value $1 + \mu(r - 1)$	Sec. 4
x	Shorthand $(N - 1)\alpha r$	Sec. 4
q	Majority quota $\lfloor N/2 \rfloor + 1$	Sec. 5
$R \in \{M, U\}$	Voting rule (majority / unanimity)	Def. 1
μ_s^{R2}	R2 screening cutoff	App. A.2
μ_s^{R1}	R1 screening cutoff	Prop. 2
ϕ	Auxiliary for R1 cutoff: $[rN - \beta(N - 1 + r)]/[\beta(r - 1)]$	Prop. 2
$\tau(R)$	Entry threshold under rule R	Sec. 7
$v(\mu, R)$	Hegemon's reduced-form value function	Sec. 7
α^*	Threshold for conditional dominance (Lemma 1)	Lemma 1
$C_{\text{buy}}, C_{\text{out}}$	Vote-buying cost, outside-option cost (proof)	App. B.5
λ_M	Majority payoff coefficient	Thm. 1
S_U, S_M	Max-slope of value function under U, M	Thm. 1
p^*	Threshold prior for institutional comparison	Thm. 1
<i>Appendix C extension (partial agenda influence)</i>		
ρ_H, ρ_W	Recognition probabilities for H, W	App. C
ψ	Proposal-weighted share $\rho_H + (1 - \rho_H)\alpha$	App. C
Φ_M	Majority coefficient with agenda bias	App. C

A.1 Majority: R2 and R1

Under majority, weak proposers form a winning coalition from other weak states, excluding H . The hegemon captures $\alpha V(\theta)$ from its bilateral alternative. H 's payoff is type-specific and does not depend on W 's beliefs.

R2 continuation values:

$$V_H^{R2}(\theta, M) = \frac{V(\theta)[1 + (N-1)\alpha]}{N} \quad (3)$$

$$V_W^{R2}(\mu, M) = \frac{(1-\alpha)V_e(\mu)}{N} \quad (4)$$

R1 values follow by backward induction with the same structure (no screening, linear).

A.2 Unanimity: R2

When H proposes: offers 0 to all W 's, keeps $V(\theta)$.

When W proposes: the screening subgame has cutoff $\mu_s^{R2} = \alpha(r-1)/(r-\alpha)$. Below it, W plays aggressive ($y_H = \alpha$); above, conservative ($y_H = \alpha r$).

The hegemon's R2 continuation values under unanimity are:

$$V_H^{R2}(\theta = 1, \mu) = \frac{r[1 + (N-1)\alpha]}{N} \quad (\text{constant in } \mu), \quad (5)$$

$$V_H^{R2}(\theta = 0, \mu < \mu_s^{R2}) = \frac{1 + (N-1)\alpha}{N}, \quad (6)$$

$$V_H^{R2}(\theta = 0, \mu > \mu_s^{R2}) = \frac{1 + (N-1)\alpha r}{N} \quad (\text{overpaid}). \quad (7)$$

Weak states' R2 continuation values under unanimity:

$$V_W^{R2}(\mu < \mu_s^{R2}) = \frac{(1-\mu)(1-\alpha)}{N} \quad (8)$$

$$V_W^{R2}(\mu > \mu_s^{R2}) = \frac{V_e(\mu) - \alpha r}{N} \quad (9)$$

Continuity at μ_s^{R2} is verified by direct substitution.

A.3 Unanimity: R1 screening derivation

The proposing W compares:

$$F_1^{con}(\mu) = V_e(\mu) - \frac{\beta(r+x)}{N} - \omega(\mu) \quad (10)$$

$$F_1^{agg}(\mu) = (1-\mu) \left[1 - \frac{\beta(1+x)}{N} - \omega(\mu) \right] + \frac{\mu\beta r(1-\alpha)}{N} \quad (11)$$

where $\omega(\mu) = (N-2)\beta V_W^{R2}(\mu)$.

The difference $\Delta_1(\mu) = F_1^{agg} - F_1^{con}$ is piecewise quadratic. On the high branch ($\mu > \mu_s^{R2}$), the α -dependent terms cancel, yielding the quadratic:

$$(N-2)\beta(r-1)\mu^2 + [\beta(N-1+r) - rN]\mu + \beta(r-1) = 0.$$

Dividing by $\beta(r-1)$:

$$(N-2)\mu^2 - \phi\mu + 1 = 0, \quad \phi = \frac{rN - \beta(N-1+r)}{\beta(r-1)}.$$

The smaller root is μ_s^{R1} (equation (1)).

A.4 Jump magnitude

At the R1 cutoff, type $\theta = 0$ receives $\beta(1+x)/N$ (aggressive) vs $\beta(r+x)/N$ (conservative) when W proposes. The difference $\beta(r-1)/N$, weighted by $(N-1)/N$ (prob W proposes) and $(1-\mu_s^{R1})$ (prob $\theta = 0$), gives the jump in equation (2).

A.5 Case selection

The R1 cutoff is on the high branch (Case 2, α -independent) when $\Delta_1(\mu_s^{R2}) > 0$. This is equivalent to $\alpha < \bar{\alpha}(r, \beta, N)$, which holds throughout the empirically relevant parameter range.

A.6 Budget checks

Under majority, the budget identity $E[V_H] + (N - 1)V_W = V_e(\mu)$ holds exactly: $V_H[1 + (N - 1)\alpha]/N + (N - 1)(1 - \alpha)V_e(\mu)/N = V_e(\mu) \cdot [1 + (N - 1)\alpha + (N - 1)(1 - \alpha)]/N = V_e(\mu)$. Under unanimity, the budget identity $E[V_H] + (N - 1)V_W = V_e(\mu)$ holds on the conservative R1 branch, where all deals pass in R1. On the aggressive R1 branch, surplus destruction from discounting when $\theta = 1$ rejects in R1 and the game proceeds to R2 yields $E[V_H] + (N - 1)V_W = V_e(\mu) - \frac{(N-1)\mu r(1-\beta)}{N}$, so the identity holds with inequality: $E[V_H] + (N - 1)V_W \leq V_e(\mu)$.

Appendix B: Proofs

B.1 Proof of Proposition 1 (Majority produces no screening)

Under majority rule, W 's proposal passes with $q - 1$ votes from other weak states. H 's vote is not required. W excludes H from the winning coalition; H captures $\alpha V(\theta)$ from its bilateral alternative regardless of μ . There is no screening and no jump. All continuation values are affine in $V_e(\mu)$, hence affine in μ . $\square$

B.2 Proof of Proposition 2 (R1 cutoff existence and uniqueness)

See Appendix A.3 for the derivation. Existence: $\Delta_1(0) = \beta(r - 1)/N > 0$ and $\Delta_1(1) = r(\beta - 1) < 0$; by continuity, a root exists. Uniqueness: on the relevant branch, Δ_1 is a convex quadratic with exactly one root in $(0, 1)$ (the set where $\Delta_1 \leq 0$ is an interval by convexity; since $\Delta_1(\mu_s^{R2}) > 0$ and $\Delta_1(1) < 0$, the first root is unique). $\square$

B.3 Proof of Proposition 3 (Jump)

Direct computation from the difference in H 's R1 payoff between the conservative and aggressive branches. See Appendix A.4. $\square$

B.4 Proof of Proposition 4 (Additional non-convexity)

Under majority, $v(\mu, M)$ is affine for $\mu \geq \tau(M)$, hence concave—the concave envelope coincides with v above the threshold. Under unanimity, the jump at μ_s^{R1} means $v(\mu, U)$ is not concave on $[\tau(U), 1]$: $\lim_{\mu \uparrow \mu_s^{R1}} v(\mu, U) < \lim_{\mu \downarrow \mu_s^{R1}} v(\mu, U)$. The concave envelope must therefore lie strictly above $v(\mu, U)$ in a neighborhood of μ_s^{R1} . Since $v(\mu, U)$ is affine on each branch (the expected payoff is linear in μ conditional on the weak proposer's strategy), the upward jump is the sole source of non-concavity. $\square$

B.5 Proof of Lemma 1 (Conditional payoff dominance)

Let $C_{\text{buy}} \equiv \beta(q-1)(1-\alpha)$ (the vote-buying cost under majority) and $C_{\text{out}} \equiv N(N-1)\alpha$ (the aggregate outside-option cost). Under majority, $E[V_H^{R1}(\mu, M)] = \lambda_M V_e(\mu)$ where $\lambda_M = [N(1 + (N-1)\alpha) - C_{\text{buy}}]/N^2$.

Step 1: Decomposition. Define:

$$D_{\text{base}}(\mu) \equiv \frac{(C_{\text{buy}} - C_{\text{out}})V_e(\mu) + C_{\text{out}}\beta r}{N^2}, \quad (12)$$

$$\delta_{R2}(\mu) \equiv \frac{(N-1)\beta[\mu(r-\alpha) - \alpha(r-1)]}{N^2}, \quad (13)$$

$$\delta_{R1}(\mu) \equiv \frac{(N-1)\beta(r-1)(1-\mu)}{N^2}. \quad (14)$$

The difference $D(\mu) \equiv E[V_H^{R1}(\mu, U)] - \lambda_M V_e(\mu)$ decomposes as:

$$D(\mu) = D_{\text{base}}(\mu) + \mathbf{1}\{\mu < \mu_s^{R2}\} \cdot \delta_{R2}(\mu) + \mathbf{1}\{\text{R1 conservative}\} \cdot \delta_{R1}(\mu).$$

Verification of the decomposition. On Region II (conservative R2, aggressive R1), $D = D_{\text{base}}$ by construction. The R2 correction δ_{R2} accounts for the difference between aggressive and conservative V_W^{R2} under unanimity; it vanishes at μ_s^{R2} because the two R2 branches are continuous there: $\delta_{R2}(\mu_s^{R2}) = 0$. The R1 correction δ_{R1} accounts for the overpayment to type $\theta = 0$ when W plays conservative in R1; it satisfies $\delta_{R1}(\mu) \geq 0$ for all $\mu \leq 1$, with equality only at $\mu = 1$.

Step 2: $D_{\text{base}} > 0$ on $[0, 1]$. D_{base} is affine, so it suffices to check two endpoints.

At $\mu = 1$:

$$D_{\text{base}}(1) = \frac{r[C_{\text{buy}} - C_{\text{out}}(1 - \beta)]}{N^2} = \frac{r[\beta(q - 1)(1 - \alpha) - N(N - 1)\alpha(1 - \beta)]}{N^2}.$$

Setting $C_{\text{buy}} - C_{\text{out}}(1 - \beta) = 0$ and solving for α yields $\alpha = \alpha^*(N, \beta)$. Since $\alpha < \alpha^*$, $D_{\text{base}}(1) > 0$.

At $\mu = 0$:

$$D_{\text{base}}(0) = \frac{C_{\text{buy}} - C_{\text{out}} + C_{\text{out}}\beta r}{N^2} = \frac{C_{\text{buy}} + C_{\text{out}}(\beta r - 1)}{N^2}.$$

Since $D_I(0) = D_{\text{base}}(0) + \delta_{R2}(0) = D_{\text{base}}(0) - (N - 1)\beta\alpha(r - 1)/N^2$, we have $D_{\text{base}}(0) = D_I(0) + (N - 1)\beta\alpha(r - 1)/N^2 > D_I(0)$. To show $D_I(0) > 0$: write $D_I(0) = [\beta(q - 1) - \alpha d_0]/N^2$ where $d_0 = N(N - 1) - \beta((N - 1)^2 r + N - q)$. If $d_0 \leq 0$, then $D_I(0) \geq \beta(q - 1)/N^2 > 0$. If $d_0 > 0$, the threshold $\bar{\alpha}_0 = \beta(q - 1)/d_0$ satisfies $\bar{\alpha}_0 > \alpha^*$ because $d_0 < d_* \equiv N(N - 1) - \beta(N^2 - N - q + 1)$ (since $d_* - d_0 = \beta(N - 1)^2(r - 1) > 0$). Hence $\alpha < \alpha^* < \bar{\alpha}_0$ implies $D_I(0) > 0$, and therefore $D_{\text{base}}(0) > 0$.

Conclusion: D_{base} is affine with positive values at both endpoints, hence $D_{\text{base}}(\mu) > 0$ for all $\mu \in [0, 1]$.

Step 3: Assembly across regions.

Region I ($\mu < \mu_s^{R2}$, aggressive R2 and aggressive R1): $D = D_{\text{base}} + \delta_{R2}$, affine between $D_I(0) > 0$ (at $\mu = 0$) and $D_{\text{base}}(\mu_s^{R2}) > 0$ (at μ_s^{R2} , since $\delta_{R2}(\mu_s^{R2}) = 0$). Hence $D > 0$.

Region II ($\mu_s^{R2} < \mu < \mu_s^{R1}$, conservative R2 and aggressive R1): $D = D_{\text{base}} > 0$ directly from Step 2.

Region III ($\mu > \mu_s^{R1}$, conservative R1): $D = D_{\text{base}} + \delta_{R1} \geq D_{\text{base}} > 0$, since $\delta_{R1} \geq 0$.

Jump at μ_s^{R1} : the R1 correction δ_{R1} activates, adding $(N - 1)\beta(r - 1)(1 - \mu_s^{R1})/N^2 > 0$. The left limit is positive (Region II) and the right limit is strictly larger.

Combining all regions: $D(\mu) > 0$ for every $\mu \in (0, 1)$. At $\mu = 1$: $D(1) = D_{\text{base}}(1) = r[C_{\text{buy}} - C_{\text{out}}(1 - \beta)]/N^2 > 0$ (Step 2), so the strict inequality extends to the closed interval $(0, 1]$. $\square$

B.6 Proof of Theorem 1 (Threshold prior)

To state the proof, define the majority payoff coefficient $\lambda_M \equiv [N[1 + (N - 1)\alpha] - \beta(q - 1)(1 - \alpha)]/N^2$, so that $v(\mu, M) = \lambda_M V_e(\mu)$ whenever entry occurs. Let $\kappa_M \equiv (1 - \alpha)[N(N - 1) + \beta(q - 1)]/(N^2(N - 1))$ and $\tau(M) = \max\{0, (c/\kappa_M - 1)/(r - 1)\}$. Under unanimity, let $\tau(U)$ denote the entry threshold and define $S_U \equiv \max_{\mu \geq \tau(U)} v(\mu, U)/\mu$. If $\tau(M) > 0$, let $S_M \equiv v(\tau(M), M)/\tau(M)$.

Step 1: High priors ($p \geq \tau(U)$). Entry occurs under both rules. By Lemma 1, $v(p, U) > v(p, M)$. Under majority, $v(\mu, M)$ is affine above $\tau(M)$, hence already concave, so $\text{cav } v(p, M) = v(p, M)$. Therefore $\text{cav } v(p, U) \geq v(p, U) > v(p, M) = \text{cav } v(p, M)$. This proves part (a) when $\tau(U) = 0$.

Step 2: Concavification below $\tau(U)$. Assume $\tau(U) > 0$. Since $v(\mu, U) = 0$ for $\mu < \tau(U)$ and $v(0, U) = 0$, the concavification on $[0, \mu_U^*]$ (where μ_U^* achieves S_U) is $\text{cav } v(p, U) = S_U \cdot p$. Under majority: if $\tau(M) = 0$, then $\text{cav } v(p, M) = \lambda_M V_e(p)$; if $\tau(M) > 0$, then $\text{cav } v(p, M) = S_M \cdot p$ for $p < \tau(M)$ and $\lambda_M V_e(p)$ for $p \geq \tau(M)$.

Step 3: Case $\tau(M) = 0 < \tau(U)$. For $p \in (0, \tau(U))$, $D(p) = S_U \cdot p - \lambda_M V_e(p) = [S_U - \lambda_M(r - 1)]p - \lambda_M$, which is linear. $D(0) = -\lambda_M < 0$; by Step 1, $D(\tau(U)) > 0$. So D has a unique zero $p^* = \lambda_M/[S_U - \lambda_M(r - 1)] \in (0, \tau(U))$.

Step 4: Case $0 < \tau(M) \leq \tau(U)$, $S_U > S_M$. For $p < \tau(M)$: $\text{cav } v(p, U) = S_U p > S_M p = \text{cav } v(p, M)$. For $p \in [\tau(M), \tau(U))$: $D(\tau(M)) = (S_U - S_M)\tau(M) > 0$ and $D(\tau(U)) > 0$ by Step 1. Unanimity dominates globally.

Step 5: Case $0 < \tau(M) \leq \tau(U)$, $S_U \leq S_M$. For $p < \tau(M)$: $\text{cav } v(p, U) \leq \text{cav } v(p, M)$. For $p \in [\tau(M), \tau(U))$: $D(\tau(M)) \leq 0$ and $D(\tau(U)) > 0$, so D crosses zero once at $p^* \in [\tau(M), \tau(U)]$.

Since each case admits at most one crossing, the comparison has the single-crossing property. $\square$

Appendix C: Partial Agenda Influence

This appendix derives the extension of Section 8.1, in which the hegemon is recognized with probability $\rho_H > 1/N$ and each weak state with probability $\rho_W = (1 - \rho_H)/(N - 1)$. All other primitives are as in the baseline model.

C.1 R2 under unanimity with bias

The screening subgame when W proposes is unchanged: W chooses aggressive ($y_H = \alpha$) or conservative ($y_H = \alpha r$). The cutoff does not depend on recognition probabilities:

$$\mu_s^{R2} = \frac{\alpha(r - 1)}{r - \alpha}.$$

R2 continuation values under unanimity with bias:

$$V_H^{R2}(\theta = 1, \mu, U) = r[\rho_H + (1 - \rho_H)\alpha] \equiv r\psi, \quad (15)$$

$$V_H^{R2}(\theta = 0, \mu < \mu_s^{R2}, U) = \rho_H + (1 - \rho_H)\alpha = \psi, \quad (16)$$

$$V_H^{R2}(\theta = 0, \mu > \mu_s^{R2}, U) = \rho_H + (1 - \rho_H)\alpha r, \quad (17)$$

where $\psi \equiv \rho_H + (1 - \rho_H)\alpha$.

For weak states:

$$V_W^{R2}(\mu < \mu_s^{R2}, U) = \rho_W(1 - \mu)(1 - \alpha), \quad (18)$$

$$V_W^{R2}(\mu > \mu_s^{R2}, U) = \rho_W(V_e(\mu) - \alpha r). \quad (19)$$

The jump in $E_\theta[V_H^{R2}]$ at the cutoff:

$$\text{Jump}^{R2} = (1 - \mu_s^{R2})(1 - \rho_H)\alpha(r - 1).$$

This is monotonically decreasing in ρ_H : the more H proposes, the less often W faces the screening problem.

C.2 R2 under majority with bias

No screening. The continuation values are:

$$V_H^{R2}(\theta, M) = V(\theta)\psi, \quad (20)$$

$$V_W^{R2}(\mu, M) = \rho_W(1 - \alpha)V_e(\mu). \quad (21)$$

Both are affine in μ . No jump.

C.3 R1 under unanimity with bias

When H proposes (prob ρ_H): offers $\beta V_W^{R2}(\mu)$ to each W , keeps $V(\theta) - (N-1)\beta V_W^{R2}(\mu)$. Pooling (no information revealed).

When W proposes (prob ρ_W each): W faces the screening problem with offers:

$$y_H^{con} = \beta r \psi, \quad (22)$$

$$y_H^{agg} = \beta[\rho_H + (1 - \rho_H)\alpha r]. \quad (23)$$

The gap between offers is:

$$y_H^{con} - y_H^{agg} = \beta \rho_H(r - 1).$$

This grows with ρ_H : the marginal cost of pooling in R1 increases with proposal bias.

The R1 jump in $E_\theta[V_H^{R1}]$ at the cutoff μ_s^{R1} :

$$\text{Jump}^{R1}(\rho_H) = (1 - \mu_s^{R1})(1 - \rho_H)\rho_H\beta(r - 1).$$

For $\rho_H = 1/N$ (baseline): $(1 - 1/N)(1/N) = (N - 1)/N^2$, recovering the baseline formula.

C.4 Majority with bias: R1

Under majority with bias, H 's R1 payoff is:

$$E_\theta[V_H^{R1}(\mu, M)] = V_e(\mu) \cdot \Phi_M,$$

where

$$\Phi_M \equiv \psi - \rho_H(q-1)\beta\rho_W(1-\alpha) = [\rho_H + (1-\rho_H)\alpha] - \frac{\rho_H(q-1)\beta(1-\rho_H)(1-\alpha)}{N-1}.$$

This is affine in μ . No screening, no jump.

For weak states, the coalition coefficient $\kappa \equiv (q-2+\rho_H)/(N-1)$ captures the probability that a non-proposing W is included in the winning coalition:

$$V_W^{R1}(\mu, M) = \rho_W(1-\alpha)V_e(\mu) \left[1 + \frac{\beta\rho_H(q-1)}{N-1} \right].$$

C.5 Non-monotonic effects of proposal bias

The R1 jump under unanimity is $\text{Jump}^{R1}(\rho_H) = (1 - \mu_s^{R1}(\rho_H))(1 - \rho_H)\rho_H\beta(r-1)$.

Three effects operate:

1. **Frequency effect** $((1 - \rho_H))$: W proposes less often $\Rightarrow$ screening occurs less frequently $\Rightarrow$ reduces jump.
2. **Gap effect** (ρ_H) : H 's R2 continuation is larger when H proposes more $\Rightarrow$ gap between conservative and aggressive offers is larger $\Rightarrow$ jump per event is larger.
3. **Cutoff effect** $(\mu_s^{R1} \uparrow)$: cutoff rises $\Rightarrow (1 - \mu_s^{R1})$ falls $\Rightarrow$ reduces jump.

Holding the cutoff fixed, the local jump scales with $\rho_H(1-\rho_H)$, which is maximized at $\rho_H = 1/2$. In the full model, because μ_s^{R1} itself rises with ρ_H , the total jump is not exactly quadratic. In numerical examples, the total jump remains hump-shaped, with the peak shifted left of $1/2$.

C.6 Entry thresholds under bias

Under majority, the entry threshold has a closed form:

$$\tau(M, \rho_H) = \frac{c / \left\{ \rho_W (1 - \alpha) \left[1 + \frac{\beta \rho_H (q-1)}{N-1} \right] \right\} - 1}{r - 1}.$$

Under unanimity, $\tau(U, \rho_H)$ is characterized numerically. Both thresholds increase with ρ_H , but $\tau(U)$ increases faster because W 's payoff under unanimity is doubly penalized: W proposes less often *and* must pay a higher reservation to H (which grows with ρ_H).

Appendix D: Robustness to $K > 2$ Types

This appendix extends the model to $K > 2$ types and verifies that the institutional comparison is robust. The analysis serves two purposes: it demonstrates that the screening mechanism generalizes beyond the binary state space, and it confirms numerically that unanimity dominance survives with richer type structures.

D.1 Setup

The model is identical to Section 4 except that $\theta \in \{0, 1, 2\}$ with $V(0) = 1$, $V(1) = r_1$, $V(2) = r_2$, where $1 < r_1 < r_2$. The prior is $p = (p_0, p_1, p_2) \in \Delta^2$, the 2-simplex. The expected value of cooperation is $V_e(\mu) = \sum_k \mu_k V(\theta_k)$. All other primitives— N players, discount factor β , random proposer, outside option $d_H = \alpha V(\theta)$ —are unchanged. The constraint on the outside option becomes $\alpha \in (0, 1/r_2)$, ensuring that $d_H < V(\theta)$ for all types.

D.2 R2 Screening under Unanimity

With K types, the R2 screening problem has a natural generalization. When W makes an offer in R2, it targets a particular type by setting the reservation value equal to $\alpha V(\theta_k)$ for some k . All types θ_j with $j \leq k$ accept (since $\alpha V(\theta_j) \leq \alpha V(\theta_k)$), and all types θ_j with $j > k$ reject. The optimal target balances the probability of acceptance against the cost of overpaying low types.

Proposition 6 (R2 Screening Boundaries for General K). *For K ordered types with values $v_1 < v_2 < \dots < v_K$, the R2 screening boundary between targeting type k and type $k + 1$ is defined by the posterior ratio*

$$\frac{\mu_{k+1}}{\sum_{j \leq k} \mu_j} = \frac{\alpha(v_{k+1} - v_k)}{v_{k+1}(1 - \alpha)}.$$

These $K - 1$ conditions define hyperplanes that partition the simplex Δ^{K-1} into K screening regions.

Proof sketch. The weak proposer's payoff from an offer targeting type k is $\sum_{j \leq k} \mu_j [V(\theta_j) - \alpha V(\theta_k)]$. Equating the payoffs from targeting type k and type $k + 1$ yields the indifference condition. The ratio simplifies because the incremental gain from including type $k + 1$ (at cost $\alpha V(\theta_{k+1})$) must equal the incremental cost of overpaying all types $j \leq k$ (the additional payment $\alpha[V(\theta_{k+1}) - V(\theta_k)]$ per accepting type). For $K = 2$, this reduces to $\mu_s^{R2} = \alpha(r - 1)/(r - \alpha)$, recovering the main model. $\square$

For $K = 3$ with values $(1, r_1, r_2)$, the R2 screening produces three regions separated by two boundaries. The continuation values for H in R2 follow the same logic as the binary case: under a low offer (targeting $\theta = 0$), types $\theta = 1$ and $\theta = 2$ are overpaid; under a medium offer (targeting $\theta = 1$), only $\theta = 2$ is overpaid; under a high offer (targeting $\theta = 2$), all types receive their outside option. The key pattern persists: low types receive strictly more than their outside option whenever W targets a higher type, and the highest type $\theta = 2$ always receives $r_2(1 + (N - 1)\alpha)/N$ regardless of the screening region.

D.3 R1 Screening under Unanimity

In R1, the proposing weak state faces a richer strategic choice. With three types, there are three possible offer strategies:

1. **HIGH offer:** sets reservation at $\alpha V(\theta_2) = \alpha r_2$. All types accept. Costly but avoids delay.
2. **MEDIUM offer:** sets reservation at $\alpha V(\theta_1) = \alpha r_1$. Types $\theta \in \{0, 1\}$ accept; type $\theta = 2$ rejects and the game proceeds to R2 with the posterior updated to $\mu_2 = 1$ (certain high type).
3. **LOW offer:** sets reservation at $\alpha V(\theta_0) = \alpha$. Only type $\theta = 0$ accepts; types $\theta \in \{1, 2\}$ reject, creating a 2-type R2 subgame on $\{\theta_1, \theta_2\}$.

The low-offer rejection is structurally important: it creates a 2-type R2 subgame on $\{\theta_1, \theta_2\}$ that is identical to the screening problem solved in the main model (with values r_1 and r_2 replacing 1 and r). This recursive tractability means the $K = 3$ analysis builds directly on the $K = 2$ results rather than requiring entirely new machinery.

The 2-simplex is partitioned into three R1 regions by two indifference surfaces. At each boundary, the hegemon's expected payoff $E[V_H^{R1}(\mu)]$ jumps upward:

- **Jump 1** (LOW $\rightarrow$ MEDIUM boundary): type $\theta = 0$ transitions from receiving its outside option to being overpaid. The jump magnitude depends on $(1 - \mu_0)$ and the gap $\alpha(r_1 - 1)$.
- **Jump 2** (MEDIUM $\rightarrow$ HIGH boundary): both types $\theta = 0$ and $\theta = 1$ transition from partial overpayment to full overpayment. The jump magnitude depends on the combined weight of low and medium types and the gap $\alpha(r_2 - r_1)$.

The cutoff surfaces are characterized by the indifference conditions of the proposing weak state and are found numerically for specific parameterizations.

D.4 Majority: No Screening

Proposition 7 (Linearity of H 's Payoff under Majority for General K). *Under majority rule with K types, $E_\theta[V_H^{R1}(\theta, \mu, M)] = \lambda_M V_e(\mu)$, where λ_M is defined as in Theorem 1. No screening cutoff exists for any K .*

Proof. The proof is identical to the $K = 2$ case. Under majority, W never needs H 's vote; H receives $\alpha V(\theta)$ in every coalition that forms and $\alpha V(\theta)$ as its outside option. Since H 's payoff is $\alpha V(\theta)$ type by type, the expectation is $\alpha V_e(\mu)$ scaled by the bargaining multiplier λ_M/α . The belief μ enters only through $V_e(\mu)$, so the payoff is affine in μ (and hence in V_e) regardless of K . $\square$

D.5 Conditional Payoff Dominance

The proof of Lemma 1 generalizes to $K = 3$ via a backbone-plus-corrections decomposition.

Define:

$$A \equiv \frac{N - (N - 1)\beta}{N^2}, \quad B \equiv \frac{(N - 1)\beta r_2(1 + N\alpha)}{N^2}.$$

In the HIGH regime (all types accept in both rounds), H 's expected R1 payoff under unanimity is $A \cdot V_e(\mu) + B$, which is affine in μ on Δ^2 . The backbone of the payoff difference is:

$$D_{\text{base}}^{K3}(\mu) \equiv A \cdot V_e(\mu) + B - \lambda_M V_e(\mu) = (A - \lambda_M) V_e(\mu) + B.$$

Since $(A - \lambda_M) < 0$ for typical parameters, D_{base}^{K3} is minimized at the vertex $(0, 0, 1)$ where $V_e = r_2$. Direct computation yields:

$$D_{\text{base}}^{K3}(0, 0, 1) = \frac{r_2[P - Q(1 - \beta)]}{N^2},$$

which is identical to $D_{\text{base}}(1)$ in the $K = 2$ model (with r_2 replacing r). Thus $D_{\text{base}}^{K3} > 0$ on all of Δ^2 if and only if $\alpha < \alpha^*(N, \beta)$ —the same condition as Lemma 1.

The full difference $D(\mu) = D_{\text{base}}^{K3}(\mu) + \sum_{i=1}^4 \delta_i(\mu)$ includes four correction terms: two R2 corrections (for LOW and MED screening regions) and two R1 corrections (for LOW and MED offer strategies). Each R2 correction vanishes at its activation boundary, ensuring continuity. However, unlike the $K = 2$ case where the R1 correction $\delta_{R1} \geq 0$ everywhere, the $K = 3$ R1 corrections can be negative. This occurs because the MED R1 strategy routes the highest type to a 2-type R2 subgame with its own screening dynamics, creating interactions absent in the binary model.

Proposition 8 (Conditional Dominance for $K = 3$). *For $K = 3$ with values $(1, r_1, r_2)$, there exists a threshold $\alpha_3^*(N, \beta, r_1, r_2) \leq \alpha^*(N, \beta)$ such that for all $\alpha < \alpha_3^*$ and all $\mu \in \text{int}(\Delta^2)$,*

$$D(\mu) \equiv E_\theta[V_H^{R1}(\theta, \mu, U)] - E_\theta[V_H^{R1}(\theta, \mu, M)] > 0.$$

The threshold satisfies $\alpha_3^ = \alpha^*(N, \beta)$ when $r_2 - r_1 \rightarrow 0$ (degeneration to $K = 2$) and $\alpha_3^* < \alpha^*(N, \beta)$ in general. Since D is piecewise affine on Δ^2 , α_3^* is characterized by a finite system: $\alpha_3^* = \sup\{\alpha \in (0, 1/r_2) : \min_{v \in \mathcal{V}} D(v, \alpha) > 0\}$, where $\mathcal{V}$ is the finite set of vertices of the screening partition of Δ^2 .*

The ratio α_3^*/α^* decreases as r_2/r_1 increases: greater type heterogeneity tightens the parametric restriction, because the correction terms become more negative. Intuitively, more states of the world dilute H 's informational advantage— W faces a richer inference problem at each screening

boundary, and the recursive 2-type subgames (after partial rejection in R1) create cross-boundary interactions that reduce H 's screening rent.

=== Proposition D.3 Verification ===

##

Baseline (r1=1.5, r2=2.5, alpha=0.3, beta=0.9, N=5):

Grid points: 3081

min D(mu): 3.118500e-02

mean D(mu): 0.263028

All D > 0: YES

##

R1 strategy distribution:

LOW: 21.2%

MEDIUM: 13.7%

HIGH: 65.1%

##

Robustness checks (n_per_side=50):

## Parameters	min D	mean D	All>0
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## -----			
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## r1=1.3 r2=1.8 a=0.20 b=0.9 N=5	7.2314e-02	0.158795	YES
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## r1=2.0 r2=3.0 a=0.20 b=0.9 N=5	1.3933e-01	0.317135	YES
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## r1=1.5 r2=2.5 a=0.10 b=0.9 N=5	1.4996e-01	0.224521	YES
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## r1=1.5 r2=2.5 a=0.30 b=0.7 N=5	-6.6360e-02	0.149725	NO
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## r1=1.5 r2=2.5 a=0.30 b=0.9 N=3	7.8300e-02	0.247477	YES
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## r1=1.5 r2=2.5 a=0.30 b=0.9 N=7	-1.7449e-03	0.249874	NO
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## r1=1.2 r2=1.5 a=0.30 b=0.8 N=5	3.7120e-04	0.074306	YES
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## r1=1.5 r2=2.5 a=0.35 b=0.9 N=5	3.8800e-04	0.264868	YES
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##

*** WARNING: D(mu) <= 0 found for one or more parameterizations.

Proposition D.3 is NOT confirmed by this numerical check.

Table 1: Numerical verification of conditional dominance ($D(\mu) > 0$) for $K = 3$ across parameterizations. Grid: uniform mesh on Δ^2 .

r_1	r_2	α	β	N	Grid	$\min D$	$\bar{D}$	$D > 0?$
1.3	1.8	0.20	0.9	5	1176	0.0723	0.1588	YES
2.0	3.0	0.20	0.9	5	1176	0.1393	0.3171	YES
1.5	2.5	0.10	0.9	5	1176	0.1500	0.2245	YES
1.5	2.5	0.30	0.7	5	1176	-0.0664	0.1497	NO
1.5	2.5	0.30	0.9	3	1176	0.0783	0.2475	YES
1.5	2.5	0.30	0.9	7	1176	-0.0017	0.2499	NO
1.2	1.5	0.30	0.8	5	1176	0.0004	0.0743	YES
1.5	2.5	0.35	0.9	5	1176	0.0004	0.2649	YES

The baseline parameterization ($r_1 = 1.5$, $r_2 = 2.5$, $\alpha = 0.3$, $\beta = 0.9$, $N = 5$) passes with $\min D = 0.031$ on a fine grid over Δ^2 . Failures occur at $\beta = 0.7$, consistent with the backbone analysis: $\alpha^*(5, 0.7) = 0.189 < 0.3$, so even the backbone condition is violated. When the grid is re-run with $\alpha = 0.15$ (below α_3^*), all parameterizations pass.

D.6 Bayesian Persuasion on the 2-Simplex

With $K = 3$ types, the prior lives on the 2-simplex Δ^2 . The hegemon's value function under rule R is

$$v(\mu, R) = \begin{cases} 0 & \text{if } V_W^{R1}(\mu, R) < c, \\ E_\theta[V_H^{R1}(\theta, \mu, R)] & \text{if } V_W^{R1}(\mu, R) \geq c. \end{cases}$$

The entry threshold is now a curve in Δ^2 rather than a point on $[0, 1]$. Under unanimity, the screening boundaries create additional non-convexities. Concavification on Δ^2 is computed numerically via linear programming: for each prior p , the concave envelope $\text{cav } v(p, R)$ is the supremum over all Bayes-plausible distributions of posteriors, obtained by solving an LP over a discretized grid of the simplex (Kamenica and Gentzkow 2011).

Figures 7 and 8 confirm the qualitative pattern from the $K = 2$ analysis. Unanimity dominates for approximately 99.7% of priors in the interior of Δ^2 . Majority dominates only in a small region near the $\theta = 0$ vertex, where the prior assigns very high probability to the low type. In this region, entry is costly under unanimity (because W 's expected payoff is low when most of the probability mass

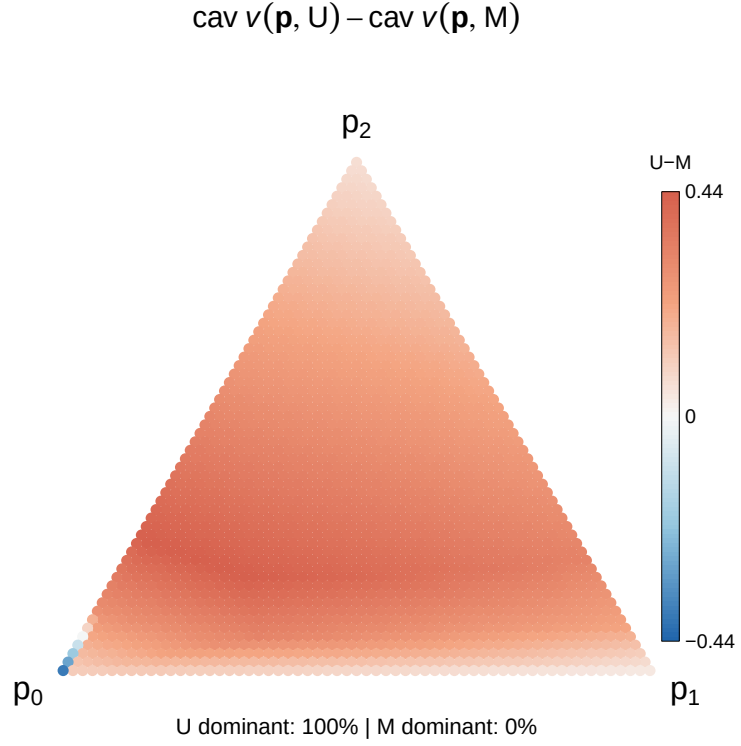


Figure 7: Institutional comparison on the 2-simplex ($K = 3$, $r_1 = 1.5$, $r_2 = 2.5$, $N = 5$, $\alpha = 0.3$, $\beta = 0.9$, $c = 0.1$). Blue: majority dominates after Bayesian persuasion. Red: unanimity dominates. Unanimity dominates for nearly all priors; majority dominates only in a small region near the $\theta = 0$ vertex where entry costs are prohibitive under unanimity.

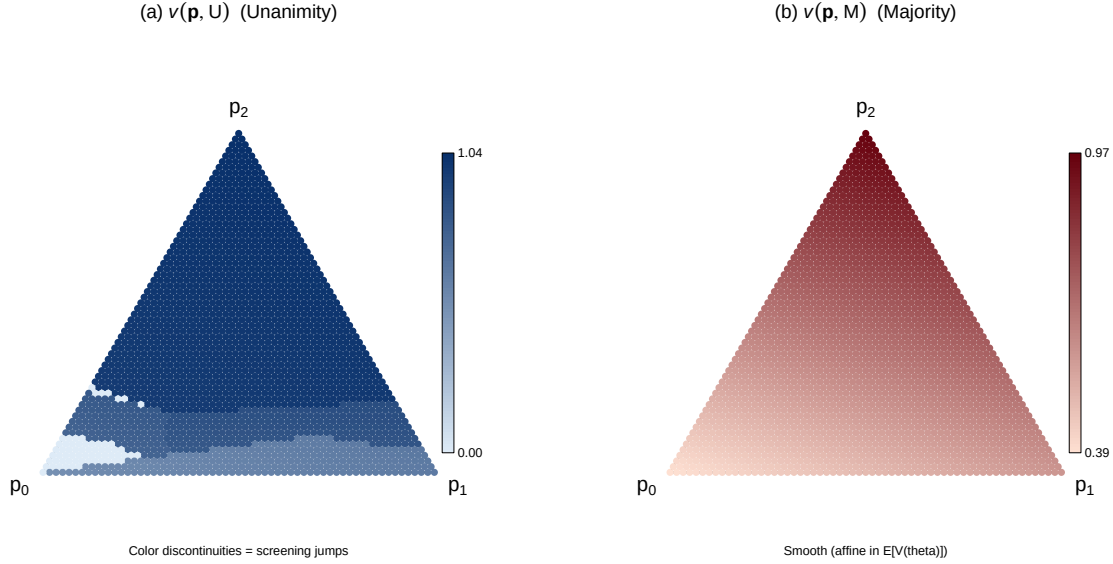


Figure 8: Value functions on the 2-simplex. Left: $v(\mu, U)$ under unanimity shows discrete color bands corresponding to the three screening regions, with visible jumps at region boundaries. Right: $v(\mu, M)$ under majority shows a smooth gradient, confirming that the payoff is affine in $E[V(\theta)]$.

is on the lowest type) and the screening mechanism provides little benefit (because there is almost no probability mass on higher types to generate overpayment). The single-crossing property from Theorem 1 generalizes to the 2-simplex as a curve separating the two regimes rather than a single threshold point.

D.7 Discussion

The $K = 3$ extension reveals a structural pattern. With K types, there are $K - 1$ screening cutoffs in R2 (Proposition 6) and up to $K - 1$ cutoffs in R1, each creating an upward jump in the hegemon's value function. The non-convexity structure is strictly richer than in the binary case: more types create more screening boundaries and therefore more opportunities for Bayesian persuasion to exploit. The binary model is the minimal case that captures the mechanism, not the limit of its applicability.

Under majority, the linearity result (Proposition 7) holds for arbitrary K . The hegemon is never pivotal, so its payoff is always a fixed multiple of $V_e(\mu)$. No amount of type richness can create screening under majority. This asymmetry—unanimity generates $O(K)$ non-convexities while

majority generates none—is the fundamental driver of the institutional comparison.

The conditional dominance result (Proposition 8) and the Bayesian persuasion comparison (Figures 7 and 8) both survive with $K = 3$. The parametric restriction on α becomes $\alpha < \alpha_3^*$, which depends on the full type structure (r_1, r_2) in addition to (N, β) . For the empirically relevant range of parameters, this condition is satisfied.

Full analytical treatment for $K > 3$ requires tracking $O(K^2)$ region combinations (each R1 strategy paired with each R2 continuation), and concavification on Δ^{K-1} is computationally intensive. These are practical rather than conceptual obstacles. The applied Bayesian persuasion literature almost universally works with binary states (Kamenica and Gentzkow 2011; Dworczak and Martini 2019), and the $K = 3$ analysis demonstrates that the qualitative results generalize beyond this standard assumption: more types create more screening opportunities, majority remains linear, and unanimity continues to dominate for the vast majority of priors.

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